MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

THIRD SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG3DSCSTA203- Statistical Research Techniques Using Softwares

Duration: 1Hour **Maximum: 30 Marks**

PRACTICAL EXAMINATION

From the following eight questions, students must answer any **FIVE**

questions. Each question carries **SIX** marks.

Solve the questions using R/spreadsheet/Python in computer lab . The data in the question along with their answer must be written in the answer paper

1. Students are allowed to join primary schools only after completing 5 years of age. It is expected that if the rule is followed, the average age of a child at the time of joining schools will be 5.4 years with sd 0.5 years. To examine whether there is any truth in the complaint that students of lower ages are also admitted to schools, a random sample of 100 students was taken and their average age at the time of joining schools was found to be 5.2 years. Is the complaint likely to be correct? [A] [5]
2. In a random sample of size 500, the mean is found to be 20. In another independent sample of size 400, the mean is 15. Could the samples have been drawn from a population with S.D 4? (CO 5) [A] [5]
3. A certain injection administered to each of 12 patients resulted in the following increases of blood pressure : 5, 2, 8, -1, 3, 0, 6, -2, 1, 5, 0, 4. can it be concluded that the injection will be, in general, accompanied by an increase in B.P? [A] [5]
4. The following are the average weekly losses of working hours due to accidents in 10 industrial plants before and after an introduction of a safety program was put into operation.

Before	45	73	46	124	33	57	83	34	26	17
After	36	60	44	119	35	51	77	29	24	11

Use 0.05 level of significance to test whether the safety program is effective.

[A] [5]

5. The following are the values of cephalic index found in two samples of skulls, one consisting of 15 and the other 13 individuals:

Sample 1: 74.1 77.7 74.0 74.4 73.8 79.3 75.8 82.8 72.2 75.2 78.2 77.1 78.4 76.3 76.8

Sample 2: 70.8 74.9 74.2 40.4 69.2 72.2 76.8 72.4 77.4 78.1 72.8 74.3 74.7

(a) Test the hypothesis that the means of the two populations are equal.

(b) Is it possible that the sample 2 has come from a population with mean 72.0.

[An] [5]

6. Manures M1, M2, M3, M4 and M5 are applied to 4 plots each in 20 identical plots in which the same variety of wheat was cultivated. The following table gives the yields per plot in quintals. Use Analysis of Variance, test whether the manures have significantly different effects at 5% level of significance.

M1	6	8	4	2
M2	7	5	5	3
M3	4	7	3	2
M4	8	4	6	7
M5	2	5	4	4

[An] [5]

7. The following data gives a sample of 300 car owners who were classified with respect to age and the number of accidents they had during the past two years. Test whether there is any relationship between these two variables?

Age ⇓ Accidents ⇒	0	1 or 2	3 or more
Less than 21	8	23	14
22-26	21	42	12
27 or more	71	90	19

[An] [5]

8. The table below shows the hours of relief provided by two analgesic drugs in 12 patients suffering from arthritis. Use appropriate non-parametric test to test the claim that Drug B is more effective than Drug A.

Patient	1	2	3	4	5	6	7	8	9	10	11	12
Drug A	2.0	3.6	2.6	2.6	7.3	3.4	14.9	6.6	2.3	2.0	6.8	8.5
Drug B	3.5	5.7	2.9	2.4	9.9	3.3	16.7	6.0	3.8	4.0	9.1	20.9

[An] [5]